Solution of the recurrence relation $a_n + 3a_{n-1} - 4a_{n-2} = 0$ with $a_0 = 7$, $a_1 = -13$

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Problem

Solve the linear homogeneous recurrence

$$a_n + 3a_{n-1} - 4a_{n-2} = 0, \qquad a_0 = 7, \quad a_1 = -13.$$

Show all steps and present a closed form for a_n.

Step 1: Characteristic equation

Seek solutions of the form $a_n = r^n$. Substituting into the recurrence gives

$$r^n + 3r^{n-1} - 4r^{n-2} = 0.$$

Divide by r^{n-2} (for $r \neq 0$):

$$r^2 + 3r - 4 = 0.$$

This quadratic is the characteristic equation.

Step 2: Solve the characteristic equation

Factor the quadratic:

$$r^2 + 3r - 4 = (r + 4)(r - 1) = 0.$$

Hence the characteristic roots are

$$r_1 = -4, \qquad r_2 = 1.$$

Step 3: General solution

For two distinct roots r_1, r_2 the general solution is a linear combination of the corresponding powers:

$$a_n = A(-4)^n + B(1)^n = A(-4)^n + B,$$

where A, B are constants determined by the initial conditions.

Step 4: Use initial conditions

Apply $n = 0$ and $n = 1$:

$$a_0 = A(-4)^0 + B = A + B = 7,$$
$$a_1 = A(-4)^1 + B = -4A + B = -13.$$

Subtract the first equation from the second to eliminate B:

$$(-4A + B) - (A + B) = -13 - 7 \quad \Rightarrow \quad -5A = -20.$$

Solving gives $A = 4$. Then $B = 7 - A = 3$.

Final closed form

Therefore the solution is

$$\boxed{a_n = 4(-4)^n + 3}$$

Checks and a few terms

Verify the initial values:

$$a_0 = 4 \cdot 1 + 3 = 7, \qquad a_1 = 4(-4) + 3 = -16 + 3 = -13.$$

Compute the next few terms using the closed form or recurrence:

$$a_2 = 4(-4)^2 + 3 = 4 \cdot 16 + 3 = 67,$$
$$a_3 = 4(-4)^3 + 3 = 4(-64) + 3 = -253.$$

A brief substitution check in the recurrence for $n = 2$:

$$a_2 + 3a_1 - 4a_0 = 67 + 3(-13) - 4(7) = 67 - 39 - 28 = 0,$$

so the recurrence is satisfied.

Summary

The unique solution satisfying $a_0 = 7, a_1 = -13$ is

$$a_n = 4(-4)^n + 3.$$

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